

Blackjack through Game Theory

Nash Equilibrium, Expected Value, and Strategic Optimization

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Overview

- **Blackjack as a game:** Complete game-theoretic formulation
- **Key literature:** Baldwin et al. (1956), Thorp (1961), modern research
- **Nash Equilibrium in blackjack:** Unique pure strategy equilibrium
- **Basic strategy:** Mathematically optimal play reducing house edge to 0.5%
- **Card counting:** How information changes the equilibrium
- **Expected Value:** Making decisions under uncertainty

Blackjack: Game theoretic formulation

Game definition

Players:

- Player (utility: maximize winnings)
- Dealer (fixed strategy: hit on < 17 , stand on ≥ 17)
- Nature (randomness: card deals)

Actions

- 1 Hit: Draw one more card
- 2 Stand: End hand
- 3 Push: Tie with the dealer
- 4 Split: Separate pair into two hands
- 5 Bust: Hand card above 21 (automatic loss)
- 6 Blackjack: Initial 21 (Ace + 10-value card, pays 3:2)

Key fact: Unlike poker, dealer strategy is fixed by casino rules, not optimized.

Blackjack: Game theoretic formulation

Game definition

- **Imperfect information:** Player cannot see dealer's hole card initially
- **Sequential:** Dealer follows fixed rule (hit soft 17 rule)
- **Stochastic:** Outcomes depend on card draw
- **Asymmetric Payoffs:** Player wins \$1, loses \$1 on bets

Nash Equilibrium: The unique pure strategy

Nash Equilibrium

Definition: No player can improve their payoff by unilaterally changing strategy.

Blackjack Equilibrium (continuous Blackjack model):

The Nash Equilibrium has a **threshold structure**:

- **Player strategy:** Stand at hand value ≥ 17 , Hit on ≤ 16
- **Dealer strategy:** Stand at ≥ 17 (fixed rule)

Intuition:

- Player hitting on $17+$ would *certainly bust* (exceed 21)
- Player standing on 16 or less gives dealer control
- Given dealer's fixed rule, this minimizes expected loss

Optimal strat

Definition: For every possible player hand and dealer upcard, the move that maximizes expected value.

- Computer analysis of millions of hands
- Expected value calculated for each decision
- Assuming infinite deck (deck composition stable)
- Dealer follows fixed rule: hit on < 17 , stand on ≥ 17

To keep in mind

- Dealer upcard matters (influences player bust risk)
- Hand composition matters (soft 17 vs hard 17)
- This is why **basic strategy** is more nuanced

Hard 16 vs Dealer 10

- **If player hit:** 62% chance to bust, 38% to improve
 - **If player stand:** Dealer hits on 16, likely has 17 to 21
 - **Optimal:** hit (smaller expected loss)
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- Expected Value: -0.53 per unit bet
 - Perfect basic strategy $\Rightarrow$ house edge **0.3%–0.6%**
 - Single-deck basic strategy $\Rightarrow$ house edge 0.17%
 - 6-deck basic strategy $\Rightarrow$ house edge 0.43%

Critical insight

Every deviation from basic strategy increases house edge.

Basic strategy chart: Hard hands

Reading the chart: Find player hand total on rows, dealer upcard on columns.

Hand	2	3	4	5	6	7	8	9	T	A
8	H	H	H	H	H	H	H	H	H	H
9	H	D	D	D	D	H	H	H	H	H
10	D	D	D	D	D	D	D	D	H	H
11	D	D	D	D	D	D	D	D	D	H
12	H	H	S	S	S	H	H	H	H	H
13	S	S	S	S	S	H	H	H	H	H
14	S	S	S	S	S	H	H	H	H	H
15	S	S	S	S	S	H	H	H	H	H
16	S	S	S	S	S	H	H	H	H	H
17+	S	S	S	S	S	S	S	S	S	S

Legend: H = Hit, S = Stand, D = Double Down

Key patterns:

- Always hit soft hands ≤ 18 (Ace counts as 11)
- Always split Aces and 8s
- Never split 10s and 5s

Expected Value: The core of Blackjack decisions

Mathematical definition:

$$EV = \sum_i P(\text{outcome}_i) \times \text{payoff}_i$$

Hit on 11 vs Dealer 6

- Probability hit 10-value card (4/13 in infinite deck): 30.77%
- New hand: 21 $\Rightarrow$ Win \$2
- Other cards (9/13): Improve but don't reach 21
- Average payoff: slightly positive

Basic strategy chart: EV analysis

Bust probabilities (Critical):

Hand total	Bust probability
12	31%
13	39%
14	46%
15	54%
16	62%
17+	100% (guaranteed)

- On 16, player loses 62% of the time if they hit
- But dealer also busts 42% of the time from their hand
- The decision: compare expected outcomes, not probabilities

Choose action with highest expected value

House edge: The reason why casinos always win in the long run

Definition:

$$\text{House edge} = \frac{\text{casino's expected winnings}}{\text{total bets}} \times 100\%$$

Typical house edge values

Without strat:

- Casual player: 2%–4% house edge
- On this setup, player loses \$2 per \$100 bet in long run

With perfect basic strategy

- Single deck (3:2 payout): 0.17%
- 6-deck (3:2 payout): 0.43%
- 8-deck (6:5 payout): 2.01% (much worse!)

Card counting: Exploiting information

The Hi-Lo card counting system:

- Count +1: Cards 2, 3, 4, 5, 6 (favor dealer bust)
- Count 0: Cards 7, 8, 9 (neutral)
- Count -1: Cards T, J, Q, K, A (favor player blackjack)

Why card counting works:

- High card count $\Rightarrow$ More 10s, Aces remain
- Player gets blackjack more often (pays 3:2)
- Dealer more likely to bust
- Player adjusts bet size: large bet on high count, minimum on low

Counting strat

When high cards remain, player has advantage. When low cards remain, casino has advantage.

Card counting: Exploiting information

EV with card counting:

- Positive count: Player EV becomes **+1%** to **+2%**
- This *reverses* the house edge
- Over 25 million hands: expected win of \$0.0148 per \$1 bet

Game theory implications:

- Original equilibrium (no counting) no longer applies
- New equilibrium: Casino reshuffles frequently (eliminates count information)
- Casinos use 6-8 deck shoes and reshuffle at 75% penetration

Key literature: Thorp (1961) "Beat the Dealer"

Strategic interaction: Player vs Casino

Extended game (with card counting awareness)

Casino's strategy:

- 1 Allow normal play (no reshuffle) $\Rightarrow$ Counters profit
- 2 Shuffle frequently $\Rightarrow$ Eliminate counting advantage
- 3 Eject suspected counters $\Rightarrow$ Deter counting

Player response:

- If casino reshuffles: Don't bother counting (no edge)
- If casino allows play: Count cards, gain +1% edge
- If casino ejects: Play conservatively to avoid detection

Casinos' reshuffle policies are themselves game-theoretically rational responses to card counting threat.

Strategic interaction: Player vs Casino

Nash Equilibrium (with counting awareness)

- Casino's best response: **Reshuffle at 75% penetration**
- Player's best response: **Play basic strategy (no count)**
- Outcome: Casino maintains 0.43% edge, counters cannot profit

Subgame Perfect Equilibrium:

- If player deviates and counts: Casino observes and reshuffles earlier
- If casino deviates and never reshuffles: Player counts and gains +1%
- Result: Casino reshuffles as optimal defensive strategy

Variance: Dispersion in the EV

Variance and Standard Deviation (SD)

EV vs. Actual outcomes:

- from EV projection: Lose \$0.43 per \$100 bet (long run, 1000s of hands)
- In 20 hands: easily win or lose \$20+ (tilt both ways)
- **SD**: How many outcomes fluctuate around EV

$$\text{Variance} = E[X^2] - (E[X])^2$$

For blackjack with perfect basic strategy:

- Expected value per hand: -0.0043 (negative)
- SD per hand: ≈ 1.1 (units of bet)
- In 50 hands: Actual outcome is typically $\pm \$55$ (almost guaranteed)
- Variance is *huge* in blackjack
- With only 20-30 hands, outcomes are nearly random

Variance: Why short term results are different than EV

Time horizon required:

Sample size	Prob of winning
20 hands	$\approx 45\%$
100 hands	$\approx 25\%$
1000 hands	$\approx 1\%$
10,000 hands	$\approx 0.0001\%$

Why Casinos allow Blackjack:

- Most players don't use basic strategy (lose 2%–4%)
- Even perfect players lose 0.43% per hand
- Volume compensates for low edge
- Card counters are rare ($< 0.1\%$ of players)

With proper strategy and long run, player lose at a slow rate. But in short term, anything is possible.

Blackjack: The unique casino game

Comparison with other games:

Game	House edge	Player control	Strategy possible?
Roulette	2.7%	None	No
Slots	2%–10%	None	No
Craps	1.4%	Minimal	Limited
Blackjack	0.43%	High	Yes
Poker	-%	Very High	Yes

Why Blackjack is special

- 1 **Player makes decisions:** Hit, stand, double, split based on information
- 2 **Information available:** Player's hand and dealer's upcard (one hole card)
- 3 **Mathematical optimality exists:** Basic strategy found through computation
- 4 **House edge is minimal:** Only 0.43% with perfect play

Information asymmetry in Blackjack

Information structure

Known info:

- Player two cards (public)
- Dealer upcard (visible)
- All previously dealt cards (in single deck games)

Unknown:

- Dealer's hole card (except in rare "no hole card" games)
- The exact composition of remaining deck

Game theoretic implication:

- This is a game of **incomplete information**
- Player make decisions with Bayesian beliefs about dealer's hand
- Example: Dealer shows 6
 - Player belief: 42% chance dealer has 16 (likely bust if hits)
 - Player belief: 38% chance dealer has 17-21 (player lose)

Information asymmetry in Blackjack

Rational expectations:

- Player assign probability to each possible dealer hand
- Compute expected value of HIT vs STAND
- Choose action with highest EV

Dealer's predictability:

- Dealer cannot bluff (has fixed rule)
- Player have *rational expectations* about dealer behavior
- This differs from poker (dealer optimizes)

Card counting as information:

- Counting reduces uncertainty about deck composition
- Changes player's Bayesian belief about high/low card probability
- Allows positive EV betting

Key literature in Blackjack game theory

① Baldwin, Cantey, Maisel, McDermott (1956)

- "Optimum Strategy in Blackjack"
- First rigorous mathematical analysis
- Computed basic strategy using hand analysis

② Thorp, Edward O. (1961)

- "Beat the Dealer"
- Introduced card counting systematically
- Showed how information changes equilibrium

③ Continuous Blackjack Model (Chen & Packman, 2020)

- "Continuous Blackjack: Equilibrium, Deviation & Adaptive Strategy"
- Full extensive-form game with Nash equilibrium
- Reinforcement learning applied to find strategies

Modern research:

- Formal Game-Theoretic Model (SSRN papers)
- Stochastic games with imperfect information
- Computational equilibrium finding using algorithms

Blackjack is asymmetric (dealer has fixed rule) but solvable to optimality.

From theory to action: Decision framework

Step 1: Identify player hand

- Hard hand: Ace counts as 1 (e.g., A-6 = 7, or 13 if counting Ace as 11)
- Soft hand: Ace counts as 11 without busting (e.g., A-6 = 17)

Step 2: Dealer's upcard

- Look at visible card (2–9, T, or A)
- This determines player strategy choice

Step 3: Consult basic strategy chart

- Find row (player hand) and column (dealer upcard)
- Follow recommendation: H, S, D, or P (Hit, Stand, Double, Split)

Step 4: Execute decision

- **Do not deviate based on intuition**
- Every deviation increases house edge
- Trust the math

Practical application: From theory to action

Common mistakes to avoid

- Taking insurance (bad EV, +7% house edge)
- Hitting on hard 17+ (certain bust)
- Standing on soft 17 (always hit)
- Not splitting Aces or 8s (huge EV loss)
- Never hitting on 12 vs dealer 2-3 (contrarian but optimal)
- **In the long run, even perfect play gives player -0.43% EV.**

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